Typed Epistemic State under Partial Knowledge: Six Matched-Information Mechanism Studies

Working framework draft

August 2026

Abstract

Research systems act on more than world facts: they carry partial knowledge about feasibility, failed interfaces, uncertain costs, transported certificates, unresolved questions, and representation edits. We ask whether explicitly typing and scoping that epistemic state changes decisions when competing mechanisms receive the same visible information. Six prospectively frozen exact-synthetic studies isolate one state operation at a time. Type-conditioned value-of-information planning recovers 71% of oracle utility and exceeds the identical uniform-prior planner. Scope-bound receipt reopening avoids both stale-failure lock-in and wasteful global reopening. Verification targeted at interval-dominance ambiguity yields 0.1096 mean regret versus 0.2518 for random verification at the same budget. Full-chain transport checking detects all constructed laundering cases with recall 1.000 and false-positive rate 0.000, including 68 deep splices that defeat last-hop inspection. Decision-coupled experiment selection avoids max-entropy decoys that consume 36.6% of pure-information-gain probes. Typed remint/transport improves utility when transport is available and ties re-derivation exactly in a prespecified regime where it has nothing to add. Two additional studies delimit the claim: an ideal value-of-information donor exactly closes the N1-C allocation-policy residual, and a crossover predictor is donor-absorbed on its original N2-F5B world, surviving only under misspecification. The evidence establishes mechanism isolation on frozen enumerable worlds, not deployment performance. LLM-labeled controls are deterministic proxies; chain identifiers are not cryptographic security; and no real research-interface or impossibility claim is made.

Keywords: epistemic state; partial knowledge; value of information; provenance; verification; active experimentation.

1 Introduction

A long-running research system accumulates facts about its own knowledge state: an interface is known feasible, unknown, or previously failed under a particular representation; a cost is measured or only bounded; a certificate was minted under an earlier context; an unanswered question is uncertain but may or may not affect the current decision. Flattening these states into generic booleans or confidence scores can erase information about when evidence remains applicable.

The scientific difficulty is attribution. If a typed system has more information than an untyped baseline, a positive result says little about typing itself. Q4 therefore uses matched-information worlds. Inside each family, every non-oracle arm receives the same serialized visible state and differs only in how it interprets or acts on that state. The strongest comparison gets first right of refusal, and each world contains a hostile regime designed to expose the shortcut that the proposed typing is meant to prevent.

Six families cover complementary operations: priors over unknown feasibility, scope-bound invalidation of failure receipts, verification under interval uncertainty, provenance-chain transport, decision-coupled experiment selection, and remint/transport across representation changes. Their purpose is not realism. Exact-synthetic construction lets the mechanism be isolated against enumerable truth, matched information, and paired outcomes.

The paper also includes negative evidence by design. One typed-state family loses its policy residual to an ideal VOI donor, and another candidate is donor-absorbed on its original world. These results matter because a mechanism-isolation programme that only publishes wins cannot establish that its first-refusal rules are doing work.

2 Methods

2.1 Prospective freezing and paired construction

Each study freezes its world generator, seed, arms, metrics, gates, tie-breaks, and honest terminal vocabulary before result-bearing execution. Episodes are paired where relevant so competing arms face the same underlying realization.

2.2 Matched information

Within a family, every non-oracle arm receives the identical serialized visible state. Mechanism comparisons therefore change the decision rule, not the available facts. Oracle arms are included only as upper/reference bounds and are not described as deployable baselines.

2.3 First right of refusal

The strongest donor-complete comparator appropriate to each construction is registered in advance. A donor tie or win closes the mechanism/policy claim. This rule is responsible for the N1-C and original-world N2-F5B negatives reported below.

2.4 Hostile validity gates

Worlds include a regime that should punish the shortcut under test: blind optimism, global reopening, partial chain inspection, decision-irrelevant information gain, stale carry-forward, or a typed operation that should provide no value. The expected trap must actually bite where prespecified; otherwise the world terminates invalid rather than yielding a positive mechanism claim.

2.5 Determinism and authority

The N4 studies use frozen seeds and deterministic decision paths. Canonical receipts encode result and scope. 'LLM_PROXY_HEURISTIC' labels refer to fixed heuristics, not measurements of a model. Authority strings explicitly limit the results to exact-synthetic mechanism isolation.

3 Results

3.1 Typed priors for unknown feasibility

In N4-A, type-conditioned priors are the only difference between 'ORION_TYPED_VOI' and the uniform-prior VOI ablation. Over 300 paired episodes, the typed arm has mean utility 3.291 versus

2.180 for uniform VOI and 4.612 for the full oracle, recovering about 71% of oracle utility. Blind optimistic commitment is driven to −13.619 mean utility and succeeds in only 19% of episodes, satisfying the hostile-world validity gate.

3.2 Scope-bound failure-receipt reopening

N4-B separates changes that invalidate a failure receipt from irrelevant context churn. Pooled mean utility is 3.199 for scoped reopening, 2.782 for never reopening, −7.813 for unscoped-change reopening, and −9.225 for always reopening. In the deliberately wasteful regime, always reopening collapses to about −13.406 while never reopening remains positive; the scoped mechanism retains the no-giveback property required by the protocol.

3.3 Targeted verification under interval costs

N4-C gives every active arm the same budget of four edge verifications. Interval-dominance targeting yields mean scalarized regret 0.1096, compared with 0.2518 for random verification, 0.2621 for midpoint optimization without targeted verification, 0.7755 for worst-case optimization, and 1.2679 for best-case optimization. The benefit is therefore attributed to where verification is spent, not verification volume.

3.4 Full-chain certificate transport

N4-D contains 200 honest and 200 laundering chains. The full-chain checker has recall 1.000 and false-positive rate 0.000. It detects all missing-receipt, spoofed-summary, and 68 deep-splice attacks. Last-hop inspection has zero recall on the deep-splice class. This is a construction-level provenance result; the identifiers are seeded values rather than cryptographic digests.

3.5 Decision-coupled active experiments

In N4-E, pure information gain is attracted by high-entropy facts that are irrelevant to the downstream choice. It spends 36.6% of its probes on decoys and obtains mean utility 7.121, while the decision-coupled selector spends none on decoys and reaches 9.266 under the shared stopping rule. The decoy-attraction gate is part of world validity.

3.6 Remint and transport across edits

N4-F3 compares typed remint/transport with naive carry-forward and matched-budget re-derivation. In the stale-hostile regime, naive carry-forward fails on 99.5% of episodes and has mean utility about −15.916; typed transport reaches 9.421 versus 7.157 for re-derivation. In the prespecified remint-unnecessary regime, all relevant arms tie exactly at 11.809659685355605 and the typed arm spends zero remints. A positive advantage there would have invalidated the study.

3.7 Negative first-refusal cases

N1-C finds a typed/scoped failure-state benefit over a scoping ablation, but an ideal VOI allocator given the same typed facts matches the candidate policy exactly at solve rate 0.9866. The allocation-policy residual is therefore closed. N2-F5B similarly gives a model-selection donor first refusal: candidate and donor both score 0.9948 on the original world, leaving only the narrower misspecified-world edge (0.9844 versus 0.9531) as a surviving bounded result.

4 Discussion

Across the six primary families, the common pattern is conditional rather than universal. Typing/scoping matters when it changes which uncertainty, receipt, certificate, or experiment is relevant to the decision. The matched-information ablations make that interpretation stronger than a comparison in which the typed arm simply sees richer state.

Equally important is the no-value behavior. N4-F3's exact tie shows that the mechanism does not need to win everywhere to support its intended claim; in the regime where remint/transport has no informational value, it does nothing and ties. N1-C pushes the same principle further: typed state can matter while the proposed policy adds nothing beyond an ideal donor. The method claim should stop at the state representation in that case.

Hostile controls are useful because they test whether a synthetic world actually exercises the proposed failure mode. They do not make the world realistic. A model can perform well on all six constructions and still fail on real research interfaces for reasons absent from the generators: misspecified state types, noisy provenance, nonstationary costs, strategic adversaries, or poor semantic extraction from raw evidence.

The next scientifically discriminating step is therefore transfer, not more synthetic variants of the same traps. A real-domain instantiation would need independently auditable state labels and matched-information baselines rather than silently substituting a richer system for the typed mechanism.

5 Related work and boundary

The component mechanisms have established parents. Robust optimization studies decision making under uncertain inputs and the trade-off between nominal performance and protection against uncertainty [1]. Data-provenance work formalizes where derived records originate and which source data influence them [2]. Active-learning literature studies how to choose information-gathering actions rather than passively consume labels [3]. Q4 also intersects value-of-information planning, workflow recovery, cache invalidation, and decision-aware experiment design; these ingredients are not claimed individually.

The bounded candidate contribution is the matched-information mechanism suite and the common typed/scoped-state interpretation across six distinct partial-knowledge operations, with donor refusal and hostile controls frozen as part of each construction. A submission-date literature pass must test whether a close existing benchmark or framework already owns that composition.

The paper is separate from Q2's negative-recovery methodology. Q2 asks how research outcomes are frozen, retained, and reopened; Q4 asks whether particular types/scopes of epistemic state are load-bearing inside controlled decision problems.

6 Limitations

All primary evidence is exact synthetic. The generators provide enumerable truth and cleanly typed state, which is the source of internal control and also the main external-validity limitation. Real research systems must infer whether a failure is stale, whether a certificate's scope changed, and whether an uncertainty is decision-relevant.

The LLM-labeled baselines are deterministic proxies. No conclusion about current or future language models follows. N4-D does not study cryptographic attackers; its hashes are construction identifiers and its completeness statement assumes end-to-end receipts cannot be forged consistently within the world definition.

N4-C reports scalarized regret, not full Pareto-front quality. N4-B excludes intra-episode receipt accrual. The programme has no lower-bound/impossibility theorem showing that typing is necessary for every mechanism family, and the registered real/generic-LLM comparison rung was not executed.

Finally, multiple positive synthetic families do not constitute independent populations. Their common design discipline reduces some confounds but cannot support a statistical prevalence claim such as 'typing helps in six out of six kinds of real research problem'.

7 Conclusion

On six frozen partial-knowledge constructions, explicitly typed and scoped epistemic state changes decision quality even when compared mechanisms receive the same visible information. The effect appears in priors, receipt invalidation, verification targeting, provenance-chain checking, experiment choice, and remint/transport. Prespecified hostile controls show that the corresponding untyped shortcuts are genuinely exercised, while an exact no-value tie and two donor-closure cases bound the claim.

The result is a mechanism-isolation benchmark, not a deployment result. It supports studying epistemic-state type and scope as first-class decision variables and provides deterministic worlds in which future mechanisms can be compared without conflating state content with state interpretation.

8 Reproducibility

Protocols are committed under `development/orion-q-nlane-closure/`; canonical results and generating modules are under `research/extensions/orion-q/nlanes/`. The replay ledger records deterministic re-execution of the core N-lane receipts. Each family uses a frozen seed, explicit gates, and a terminal vocabulary that distinguishes mechanism failure from world invalidity.

A reproduction should first verify the protocol/result binding, then rerun the world and all registered arms, then recompute the headline matched comparator and hostile validity gate. The negative N1-C and N2-F5B families must be verified with equal prominence. Paper-level verification steps and non-equivalences are listed in `REPRODUCE.md`.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. Because the studies construct hostile failure modes, the principal reporting risk is over-interpreting those constructions as security or deployment evidence. The manuscript therefore labels them as exact-synthetic mechanism tests and keeps cryptographic, real-adversary, and real-agent claims outside the authority boundary.

Computational work is deterministic and bounded by the frozen episode counts/enumerations. Negative and donor-absorbed results are retained rather than discarded. No result can self-authorize P10, novelty, deployment, or physical quantum claims.

References

[1] Dimitris Bertsimas and Melvyn Sim. The price of robustness. *Operations Research*, 52(1):35–53, 2004.

[2] Peter Buneman, Sanjeev Khanna, and Wang-Chiew Tan. Why and where: A characterization of data provenance. In *Database Theory—ICDT 2001*, volume 1973 of *Lecture Notes in Computer Science*, pages 316–330. Springer, 2001.

[3] Burr Settles. Active learning literature survey. Technical Report 1648, University of Wisconsin–Madison, Department of Computer Sciences, 2009.